

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: _____

PRN / Roll No: _____

Division: _____

UNIT V: BASICS OF INVESTMENT

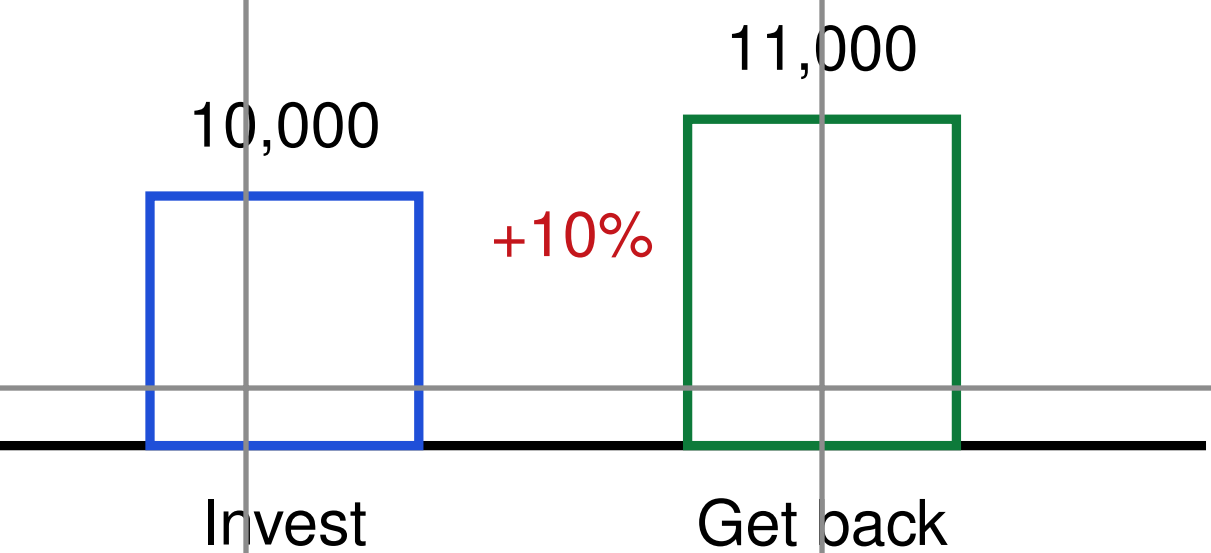
1. Investment Return

The gain (income + price rise) on an investment, as a % of the money invested.

$$\text{Return} = \frac{\text{Gain}}{\text{Invested}} \times 100$$

Eg: Invest Rs 10,000, get back Rs 11,000.

$$\text{Return} = 1,000 \div 10,000 = 10\%$$



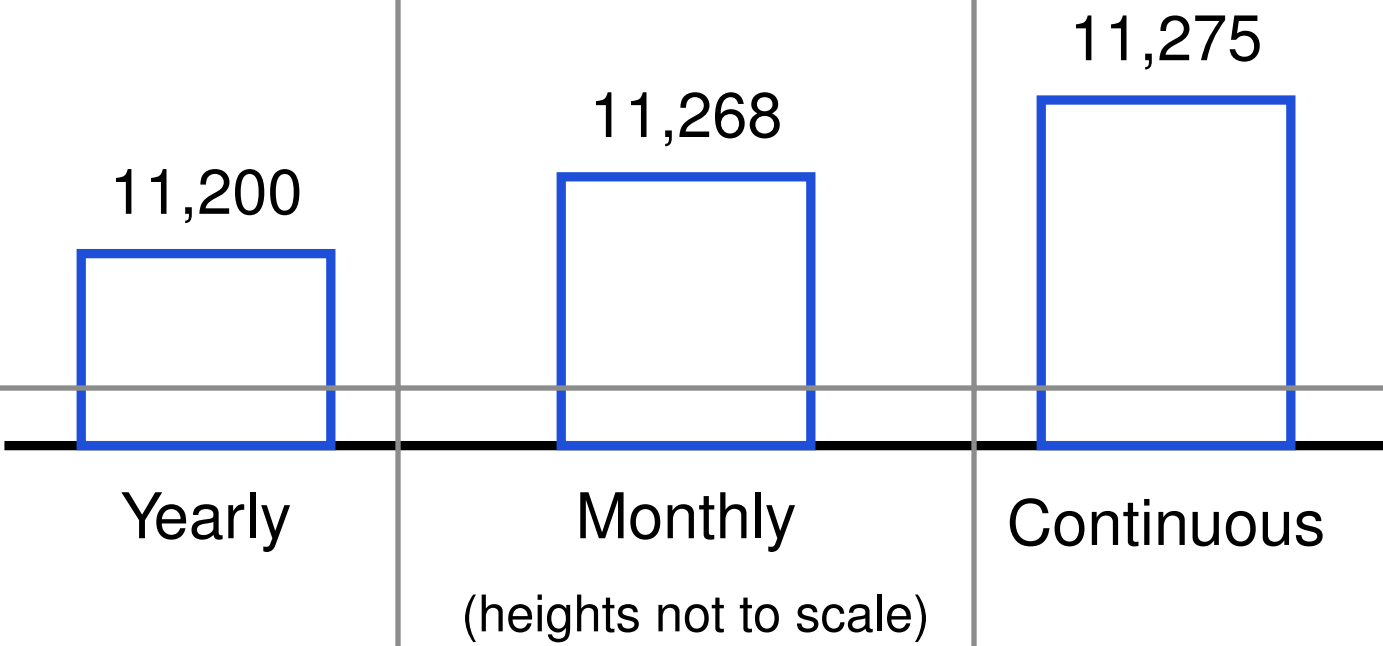
2. Compounding Frequency

At the same yearly rate, adding interest more often gives a little more money.

$$A = P \left(1 + \frac{r}{m} \right)^{mt}$$

Continuous: $A = Pe^{rt}$

Eg: Rs 10,000 at 12% for 1 year



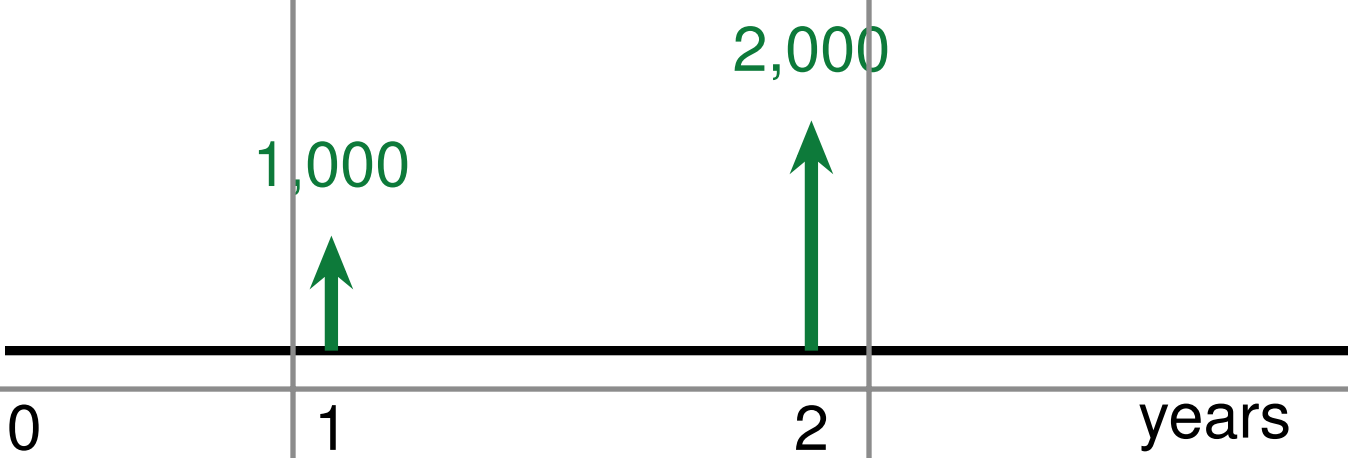
3. Uneven Cash Flows

Money received differs each year. Discount each to today and add.

$$PV = \sum \frac{CF_t}{(1+i)^t}$$

Eg: Rs 1,000 (year 1) and Rs 2,000 (year 2), $i = 10\%$

$$PV = 909 + 1,653 = \text{Rs } 2,562$$

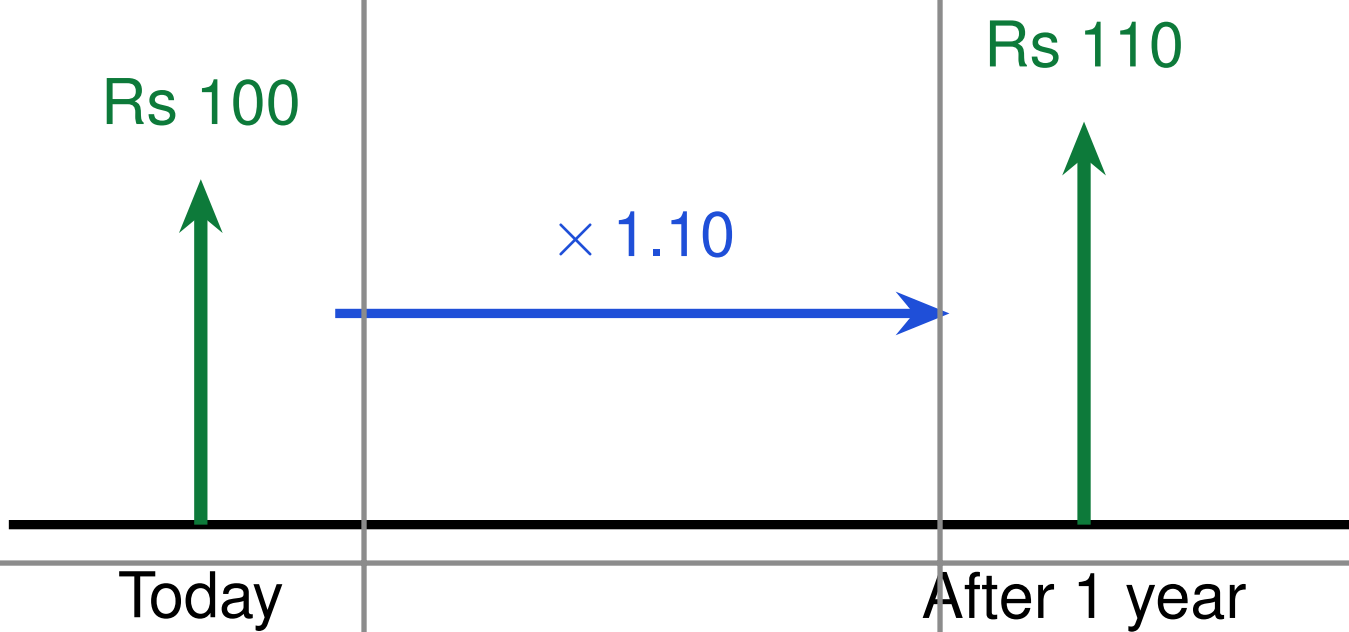


4. Economic Equivalence

At a given interest rate, different sums at different times can have the same value.

$$F = P(1+i)^n \quad P = \frac{F}{(1+i)^n}$$

Eg: at 10%, Rs 100 today = Rs 110 after 1 year



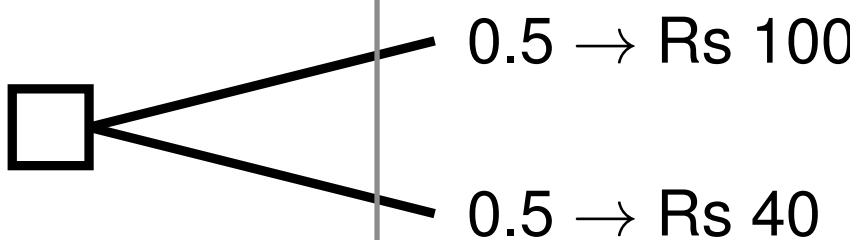
UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk

Chances are known. Choose the best average (EMV).

$$\text{EMV} = \sum (\text{chance} \times \text{payoff})$$

Eg: 50%: Rs 100, 50%: Rs 40 $\Rightarrow$ EMV = Rs 70



2. Decision under Uncertainty

Chances are unknown. Use a rule:

- **Maximax**: best of the best $\rightarrow$ **Stock**
- **Maximin**: best of the worst $\rightarrow$ **Bond**
- **Min-max regret**: least regret $\rightarrow$ **Bond**

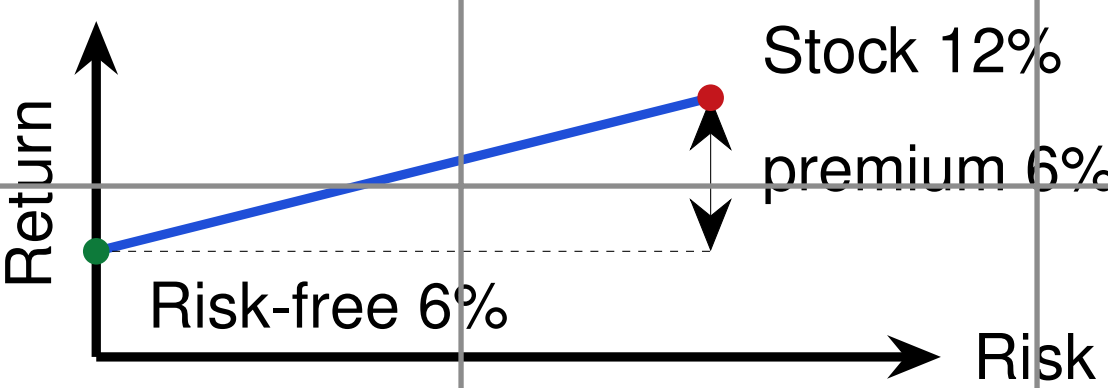
Payoff	Boom	Slump
Stock	80	-20
Bond	40	25

3. Risk Premium

The extra return a risky asset must offer over a risk-free one.

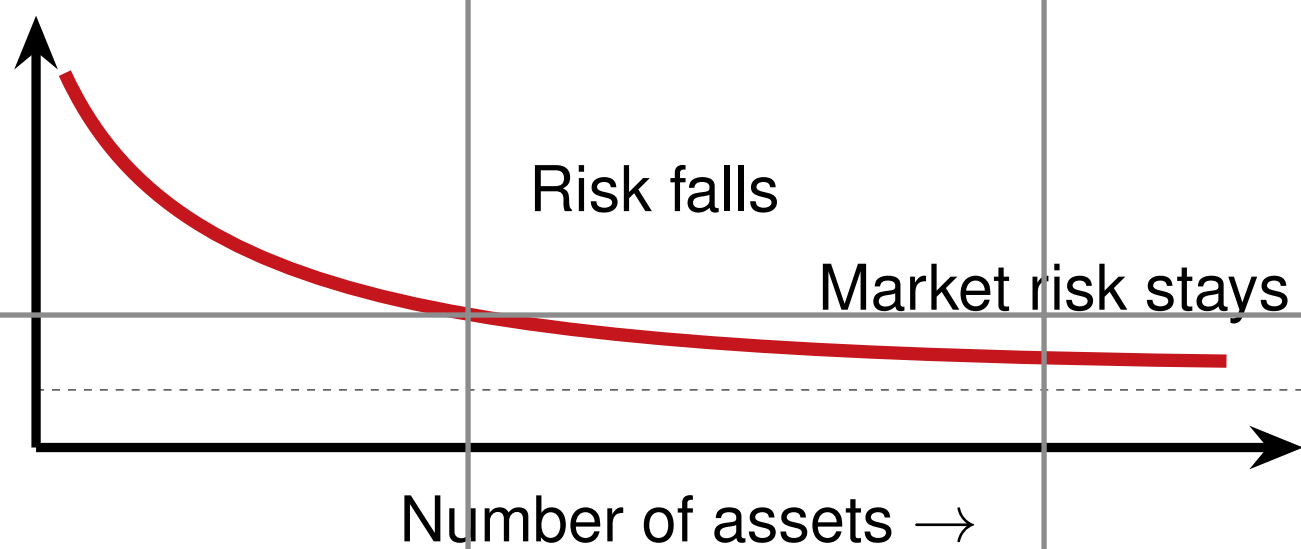
$$\text{Premium} = \text{Expected} - \text{Risk-free rate}$$

Eg: 12% - 6% = 6%



4. Portfolio Diversification

Spread money over assets that do not move together to lower risk. Market risk cannot be removed.



5. Life Insurance

You pay a premium. The insurer pays the sum assured if you die within the term.

$$\text{Premium} \approx \text{Sum assured} \times P(\text{death})$$

Eg (1 yr): Rs 1,00,000 $\times$ 0.01 = Rs 1,000



6. Endowment

Pays the sum assured if you **die in the term** OR **survive to the end**.

$$\text{Endowment} = \text{Term} + \text{Pure endowment}$$

Pure endowment = paid only if you survive.



Investors demand a higher expected return for taking higher risk